

Manual

Machine Data Management MDE-MDM 8.1

Version 1.0.8376

Last changed on: 19.06.2020

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1 Overview of Machine Data Management

Summary

Purpose

Machine data management provides functions that make it possible to automatically transfer unit quantities and machine statuses from connected machines / systems. Depending on the type of connection, there is the ability to monitor the scheduled cycle time accounting for predefined machine tolerances.

If automatic transfer to certain machines is not possible, there is the option to establish a semi-automatic connection to enter data manually.

A full range of configuration and plausibility checking functions make it possible to adapt precisely to each machine and to control the data entry process.

Implementation considerations

You use machine data management if

- You would like to transfer unit quantities and statuses that were provided from machines and systems into the system automatically.
- You would like the option to either manually enter machine statuses as they occur or have it automatically entered with the support of the system.

Integration

Machine data management is integrated into numerous other system function packages:

- Detailed scheduling / shop floor scheduling
 - The recorded machine statuses are visualized on line in shop floor scheduling on the planning board so that the scheduler is given an overview of the current production situation.
 - Depending on the system configuration, remaining run time is calculated based on the cycles or quantities recorded.
- Graphic machine park
 - In graphic machine park the recorded machine statuses and cycles / unit quantities are visualized on line.
- Tool / resource management
 - In tool / resource management, maintenance can be triggered and scheduled based on the numbers of units recorded.

Features

- Machine management
 - Database configuration tables for workplaces and machines
 - Individual configuration of up to 9,999 states/ statuses per machine or workplace; grouping of statuses into status hierarchies
 - Print function for creating a status list with barcode
 - Definition of status classes that are posted to the machine states / statuses upon occurrence
- Status recording
 - Manual recording of machine states/ statuses at BDE terminals
 - Extensive plausibility checks (e.g. check whether setup status may be set if no order is logged on)
 - Automatic recording of machine states/ statuses at BDE terminals
- Entry of quantities
 - Configuration for definition of logical counter inputs for the automatic recording of production figures
 - Automatic recording of production figures (cycles, yield and scrap) at BDE terminals
 - Manual recording of production figures (cycles, yield and scrap) at BDE terminals
- Machine monitoring
 - Monitoring of the scheduled cycle time accounting for predefined machine tolerances
 - Machine monitoring based on operating signals
 - Suppression of the production status despite active production signals (e.g. during setup status)
 - Explanation for disturbances based on machine monitoring
- Posting:
 - Posting of machine status times to machine-related bookings (postings)
 - Posting of machine status times to status classes
- Posting maintenance
 - Display of machine-related log records (postings) generated based on entered data
 - The ability to edit and correct machine-related log records

2 Data Collection in HYDRA MDE

2.1 Summary

The posting principle of HYDRA MDE corresponds to that of HYDRA BDE: once again the collection and processing of dialogs constitute the basis. Processing of a dialog results in events that constitute the basis for the updating (calculation of statuses) and posting (generation of log records).

Dialogs

As input values of data collection - dialogs represent the system's interface with the shop floor.

Events are generated by resolving the dialog. The dialog is resolved into individual events based on dialog data, the processing logic of the dialog (incl. configurations) and the application of the dialog according to the context (posting status).

HYDRA MDE input dialogs are among other things:

Dialog	Meaning
M_MST	Change (machine) status
M_SZY	Change target cycle (with monitoring based on cycles)
M_TLG	Change partitioning/cavity
M_AST	Cyclic status update is automatically generated by the terminal; cannot be recorded manually

Events

Events result from dissolving dialogs and are thus the result of the input. Which events get to the system at what times is an important point of controlling the software.

Further, partly optional, input functions are:

- logging of the dialog and the resulting events,
- provision of the posting result,
- provision of current information in the form of lists (e.g. machine lists),
- escalations (with M_MST).

3 HYDRA MDE Log Records

General

HYDRA-MDE log records describe postings based on machines/workplaces. A HYDRA-MDE log record is an evaluated data record that is generated due to posting events. An MDE log record documents among other things:

- Period of time (beginning, end) for which the log record applies,
- Duration since the last status change,
- Created status,
- Resource performance account which the status is assigned to,
- Target cycle that was set when the status changed (end of log record),
- Partitioning that was set when the status changed (end of log record),
- The number of cycles that have been recorded within the period of the log record,
- Computed quantities for meter readings that have been recorded within the period of the log record.

HYDRA-MDE log records do not have a direct relation to HYDRA-BDE log records. Thus, the period of an MDE log record does not depend on HYDRA-BDE log records. In exceptional cases, there might even be MDE log records without that an operation was logged on during that period of time (and as a result, without generating HYDRA-BDE log records for it). Which MDE log record is generated at what point in time depends, in particular, on the triggering posting event. The different log record types are distinguished by their record type. The machine-related record types that are described in the sections that follow are to be distinguished.

Record type P

General

A log record of the record type P is generated when the workplace/machine status is changed. It has been configured to be able to evaluate the period of time when a status occurred.

Triggering events and dialogs

Events: Machine status change (M_MST)

Dialog: Machine status change (M_MST),

Special remarks

The quantities included in an MDE log record are the quantities posted in this period of time. These quantities result from manual quantity postings (partial uploads A_TR) or from automatically recorded meter readings which might have been converted using conversion factors (partitioning, pulse factor).

The values in the "counter" fields do not include deltas, but quantities and cycles that have been added up since the beginning of the shift. The delta quantities (as of HYDRA-MDE 7.2) are provided in separate fields for evaluation purposes.

Record type N**General**

A log record of the record type N is generated when the shift ends (automatic shift change, requires a HYDRA-MDE machine). It has been configured to evaluate the period of time prior to the shift end when the status already existed.

Triggering events and dialogs

Events: Machine status change (M_MST)

Dialog: End of shift (A_AAB)

Special remarks

The notes given for record type "P" also apply in this context.

4 Posting of Times

The machine duration (or duration) is determined by the time interval between two status changes. It is synchronized with the shift calendar of the machine, and planned shift breaks are removed from the time interval calculation.

Please note:

The determination of durations when status messages/postings are recorded online at the terminal (or via PDM) only takes into account a limited number of shifts. This relates to the last point in time postings were made for this machine.

However, durations might be missing or incorrect in MDE log records if no postings (status changes, shift changes, or similar) are entered for a machine over a longer period of time (e.g. the terminal is shut down).

Special features

The following configuration option may influence the above-mentioned posting of times/durations:

Post production time to main utilization time (MUT) during break

Cross-system configuration in the basic parameter settings of HYDRA.

Waiting period processing, machine

Cross-system configuration in the basic parameter settings of HYDRA.

Automatic status change to status 999

Shift-related configuration at the HYDRA-BDE day type.

5 Posting of Quantities

Entry in different quantity types and quantity accounts

Quantities can be recorded in different quantity types and quantity accounts at the machine.

The following **quantity accounts** are supported:

- Yield
- Scrap
- Rework
- Open quantity (problem quantity)

The following quantity types are supported per quantity account:

- Primary quantity
- Secondary quantity
- Tertiary quantity
- Base quantity

The automatic entry of quantities at the terminal always refers to the primary quantity.

Conversion in alternative quantity units

Alternative quantity accounts may be posted as follows:

- By direct (manual) entry
- By conversion
 - From other quantity types if manually collected
 - From the primary quantity if automatically collected

Direct (manual) entry

If a quantity is directly entered (manually) in a quantity unit of a quantity type, automatic conversion will not take place.

Conversion from other quantity types

If alternative quantity accounts are not recorded, values will be converted into alternative accounts based on conversion factors or quantity units configured within the MOC master data of machines/workplaces.

In general, values are first converted into the base quantity unit (unless this one is recorded manually) and from the base quantity unit into the alternative unit (unless this one is recorded manually).

Identical quantity units

If quantity units are identical they are converted using numerator and denominator in the master data of machines/workplaces. If numerator and/or denominator = 0 quantities are taken over 1:1 without being converted.

Different quantity units

If quantity units are different, values will be converted based on the below-mentioned order of priorities:

- Conversion by means of numerator and denominator within master data of machines/workplaces (always convert into base quantity unit first, then into the alternative unit); if numerator and/or denominator = 0
- Conversion using quantity unit formulas

No quantity units

If the quantity units of the machine are not assigned, no conversion will take place.

Conversion of quantity 0

A quantity 0 is generally not converted into alternative units, even if a value that is not 0 could be calculated (e.g. by means of a formula).

Quantity conversion of automatically collected quantities

- Fixed factors/values or values based on machines/workplaces (user fields);
- Data specific to operations such as length, width, weight per piece, etc.

may have some influence on the quantity conversion of automatically recorded quantities using formulas.

As regards data specific to OPs, the operation that has been logged on for the longest time is used. Consequently, the following (logical) restriction arises for operations that are logged on at the same time when it comes to quantity conversion:

- operations must produce the same material;
- operations must have the same default data (length, width, weight per piece).

Any further requirements are to be taken into account in customer projects.

Basis for HYDRA-MDE quantity conversions

The configuration option "basis for HYDRA-MDE quantity conversion" of the configuration of workplaces/machines dialog, "configuration" tab > quantities, allows using the configured quantity conversion of running operations even for the machine. This enables a correct calculation of quantities even if more than one operation is active :

M – conversion factors of the workplace (APZ) [default]

A – conversion factors of the OP if logged on, otherwise workplace

Please note

If configuration A and different quantity units of operations are in use, quantities are accumulated on machine accounts without taking the units into account.

Quantity conversion relating to HYDRA-MDE based on data specific to operations or values based on machines/workplaces (user fields) is not supported in the maintenance of postings function. In this case, the "convert quantities" option should not be activated in the editing dialog of the maintenance of postings function.

Display of alternative quantity units at the terminal

Only Windows terminals allow for alternative quantity units to be collected (manually) and displayed (customizing required). However, it has to be taken into account that the terminal itself does not perform any local conversion into alternative quantity units. Data is only displayed by reloading data from the HYDRA server .

Automatic collection of quantities

The counters are configured per machine. The following options are available:

- Posting as yield, scrap, rework, open quantity, no posting
- Posting as cycles (strokes)
- Allocation with partitioning and/or pulse factor
- Cycle monitoring
- Reason (e.g. scrap reason)
- Allocation with "quantity account"
e.g. deducting scrap from yield

Quantities resulting from automatically recorded counters are always posted as primary quantities. The conversion into other quantity types is described in the previous section. The HYDRA server sets off quantities against other quantity accounts (e.g. when the total quantity is collected).

Counters that are configured with the "no posting" option do not post any quantity account. They may be used to represent the following fields of application, among other things:

- Cycle monitoring only
- Cycle monitoring and posting as cycles (strokes)
- Posting as cycles (strokes) without cycle monitoring

Quantity calculation and parallel make-to-order-production

For operations that are logged on simultaneously, quantities are posted with respect to the order based on the respective specifications (partitioning, pulse factor) of the operation.

All quantity accounts that are recorded automatically are determined specifically (yield, scrap, rework, open quantity). Counter pulses or quantities resulting from this are generally posted onto *all* OPs that are logged on (according to the configuration: with activated partitioning/pulse factor or not).

Specifications are determined as follows:

Partitioning/cavity

Partitioning for the machine: $(TLG_{OP1} + TLG_{OP2} + TLG_{OPn}) * TLG_{Machine}$

Partitioning for the operation: $TLG_{OP} * TLG_{Machine}$

If the last operation is interrupted or logged off, partitioning is reset to 1 and multiplied by the machine-specific partitioning.

Changing of partitioning at the terminal

If the partitioning is changed, the partitioning for the machine as well as for the operation, for which partitioning was changed, will be changed/updated as well.

Pulse factor

Pulse factor for the machine = $\text{minimum}(IMPFAKT_{OP1}, IMPFAKT_{OP2}, \dots IMPFAKT_{OPn}) * IMPFAKT_{Machine}$

Pulse factor for the OP = $\text{minimum}(IMPFAKT_{OP1}, IMPFAKT_{OP2}, \dots IMPFAKT_{OPn}) * IMPFAKT_{Machine}$

Please note:

The same pulse factor applies for all active operations. Therefore, it should be ensured that parallel operations get the same default pulse factor. This means only one pulse factor that is specific to operations is taken into account when quantities are calculated.

Quantity calculation on the basis of partitioning or pulse factor

Quantity for the machine = pulse * partitioning for the machine / pulse factor for the machine

Quantity for the operation = pulse * partitioning for the operation / pulse factor for the operation

Please note:

In terms of figures, a pulse factor is regarded as fraction. Thus, the pulse is used as the denominator for the calculation of quantities, while the partitioning is regarded as the numerator.

Display at the terminal

The factor from

partitioning for the machine / pulse factor for the machine

that is relevant for quantity calculation of the machine is displayed in the partitioning field at the terminal.

As described beforehand, the different default values of the machine (such as machine-specific partitioning) and of active OPs have some influence on this factor.

The respectively current target cycle (without consideration of the cycle extension) is displayed in the **target cycle** field at the terminal. This is $\max(\text{SZY}_{\text{OP1..n}})$ for OPs that are logged on at the same time.

Output of target quantity reached

The target quantity output of the machine interface, e.g. for activating the target quantity lamp is set as soon as the least target quantity of a registered order has been reached. In case the OP with the least target quantity is interrupted or finished, the next OP with the least target quantity is used.

Manual collection of quantities

Manually recorded quantities may additionally be posted with the following configurations:

- Set quantity (accounts) against other quantity accounts
e.g. deducting manually recorded scrap from yield
- Post manual quantities as cycles

Allocation with another quantity account

Automatically as well as manually recorded quantities may additionally be set off against another quantity account, for example, the manually recorded scrap can be deducted from yield.

The “allocation with” option is configured for:

- automatically recorded quantities within the counter configuration
- manually recorded quantities within the machine configuration

Using these options may lead to postings (HYDRA-BDE log records, HYDRA-MDE log records) with negative quantities or even to negative order quantities.

Display of decimal places at the terminal

Respective configurations are to be enabled to calculate or display quantities with decimal places.

Windows terminal configuration

Settings in ctwin.ini / ctaip.ini:

```
[MSS-INIT]
NACHKOMMA=3
```

If the NACHKOMMA parameter is not available in [MSS-INIT] HYDRA-MDE works with the setting 0.

Settings in ctwinlay.ini / ctaipplay.ini:

```
[main]
nachkommastellen=2          <- for machine overview display
```

DOS terminal configuration

Settings in ctdos.ini

```
[system]
NachkommastellenIntern=8      <- Default
NachkommastellenExtern=0     <- Default
```


6 Configuration of Workplaces and Resources

Overview

Menu	Master data → Resources → Resource configuration Master data --> Workplaces/ Machines --> Workplace configuration
Transaction code	res
Function authorization	mdres mdresgenh for fields in connection with Test Equipment Management

The resource configuration is the central function for resource management in MES.

Purpose

The master data for both workplaces and machines as well as for other resources (tools, DNC resources, etc.) are managed here. Resources are roughly classified according to the resource type. This type is also connected with corresponding functions and applications that open other functional components of the MES especially designed for the respective type.

Integration

Use this application to view the resource information of all resource types available in HYDRA. However, the editing of data records depends on the resource type. In this way, depending on the resource type, you cannot edit all fields or create and delete all resources.

Based on the resource type, other applications are present in the MES that are specially customized for these types. For example, the application package of machine data collection is based on resources of type "Machine".

In addition to the resource configuration, the "resource overview" application is available, which does not permit data maintenance, but does enable administration operations for daily handling of resources such as the stock transfer of a resource.

Requirements

Create a year model/ shift calendar prior to creating a workplace or machine. To use the various resource types in a meaningful way, purchase the advanced licenses for these types.

Selection criteria

The application provides the following selection criteria:

Resource from ... to ...

This selection criterion refers to the resource. You can also use wildcards (placeholders *).

Short name

Short name of the resource. Only relevant for resources of type MNR.

Resource type

Type of the resource.

Workplaces and machines always have the resource type MNR. But you can assign individual resource types to the other resources by configuration. Predefined resource types include:

DNC	NC-/ DNC program
DOC	Document
ENE	Energy meter
ENT	Extraction device
ENT	Extraction device
MNR	Workplace/ Machine
PAC	Packaging, transportation container
PRM	Test and measuring equipment
PER	Production staff / general
PRU	Setup staff
TEM	Tempering equipment
VOR	Device
WNR	Tool

We recommend using the predefined resource types.



The system adjusts the displayed detail resource information depending on the resource selected in the table overview.

Name

Name of the resource.

Group

Workplace/ Machine group of the resource. Only relevant for resources of type MNR.

Cost center

Cost center of the resource.

Short name

Short name of the resource

Resource family

Family to which the resource is assigned.

Responsibility area

Responsibility area to which the resource is assigned.

Storage location

Regular storage location of the resource.

MD user fields

MD user fields 1- 6 of the resource. If you select a resource family in the selection panel, the application shows the field names according to the assigned user field definition.

Field descriptions

This detail application includes four main tabs:

- Resource configuration
- Resource list
- Resource attributes
- DNC versions

"Resource configuration" tab

Here define the configurations and master data of the resources.

"General" tab**Resource type**

Resource type of the resource. When the HYDRA system is delivered, some resource types are predefined. Create additional resource types in the application [Resource types](#).

Resource

Enter the number of the resource or workplace to be collected in this field.

The maximum number of places allowed for this number is as follows and depends on the resource type:

- Resources of type MNR: maximum of 8 places
- Resources of type <> MNR: maximum of 20 places

Permitted characters include ABCDEFGHIJKLMNOPQRSTUVWXYZ1234567890/_.-+#. Do not use spaces and other special characters. For technical reasons, you can enter * (asterisk) and % (percent), but they are nonetheless not permitted because they are not valid characters. When you exit the input field, the system automatically converts lower case letters into CAPITAL LETTERS.

Please note for workplaces/ machines (resource type MNR):

For technical reasons, the system does not check the maximum number of characters allowed for resources of the type MNR. For this reason, make sure that the length of the resource number (= workplace/ machine number) does not exceed 8 characters.

Please note: You can only enter 8 characters in the GUI, if you set the resource type MNR before entering the resource ID (machine number).

If you selected the option "numeric machine number" (HYDRA basic parameter settings) for use on DOS terminals, ensure that the resource number (= workplace/ machine number) includes only numerical digits and that the length of the number is exactly 8 places. If necessary, when creating the workplace/ machine, add zeros to the beginning of the number to extend it to 8 digits.

Short name

Short name of the resource. Only use this field with workplaces/ machines (resources of the type MNR).

Name

This field is used to assign a short, unique name for each resource. This name is displayed in reports and overviews and in the terminal and it is useful for orientation.

Responsibility area

Responsibility areas are used such that in various evaluations, the user is only shown the data to which he/she has access according to his/her responsibility area authorization.

The responsibility area can also remain empty. In this case, the resource is always displayed regardless of the user's assigned responsibility authorizations.



If the responsibility area is left empty, the system automatically enters the value "-- DEFAULT--" in the field. Resources including this value are always displayed regardless of the user's assigned responsibility authorizations.

Cost center

The cost center to which the resource belongs is entered in this field.

Inventory number, engraving number, drawing number, manufacturer, owner

Additional information functioning as a comment.

Acquisition date, acquisition costs

Additional information functioning as a comment.

Configure the currency for the entire system in the HYDRA basic parameter settings.

Storage location

Location where the resource is stored when it is not being used (original storage location).

In connection with the Material and Production Logistics module (MPL), specify a material buffer in this field. If you log on an input batch, the logged on input batch(es) will be reposted from the previous material buffer to the material buffer specified in this field (upstream of the machine).

Delivery date, start-up date, guaranty date

Additional information functioning as a comment. These fields are only available if Test Equipment Management (PMV-PPK or PMV-SVP) is licensed and the right "mdresgenh" is assigned.

External designation, resource type designation, usage, purchase order number

Additional information functioning as a comment. These fields are only available if Test Equipment Management (PMV-PPK or PMV-SVP) is licensed and the right "mdresgenh" is assigned.

Supplier and party in charge including detail fields

Additional information functioning as a comment. These fields are only available if Test Equipment Management (PMV-PPK or PMV-SVP) is licensed and the right "mdresgenh" is assigned.

"Workplace configuration" tab

This tab is only available if you select a resource of the type "MNR".

Workplace master data**Workplace category**

N Machine

P Workplace

Defined as machine or workplace. As regards processing, both of these categories are identical if only BDE or MDE and PDV are in use.

J Machining center (BDE-BEA only)

The "Machining center" category and its functionality are described in detail in the BDE-BEA product documentation.

L Line (MDE-SFL only)

A Aggregate (MDE-SFL only)

The categories "Aggregate" and "Line" and their functions are described in detail in the MDE-SFL product documentation.

Q CAQ inspection station

Workplace is defined purely as a CAQ inspection station without affecting the BDE or MDE statistics.

R Reel-based manufacturing (MPL-RF only)

S Cutting unit (MPL-RF only)

The categories "Reel-based manufacturing" and "Cutting unit" and their functions are described in detail in the MPL-RF product documentation.

D Parallel output batches (MPL only, starting with MPL 7.2)

Produce parallel output batches on the machine for an operation subject to batch management.

C Packing station (MPL-PAL only, starting with MPL 7.2)

Use specific posting functions on the machine to represent a packing station. The functions are described in detail in the MPL-PAL product documentation.

M Melting aggregate

This option defines a machine as melting aggregate in terms of composition.

Workplace type

E Single workplace

G Group workplace

Please note

Terminals can be assigned to both group and single workplaces. However, in this case make sure that the terminal is set to operation mode "BDE" or the option **Processing** is set to "BDE processing" when assigning workplaces to terminals.

External workplace

This field identifies external workplaces. Currently, it only functions as a comment.

Blocked

If this identifier is set, the machine/ workplace has been (logically) deleted. In this case, the system does no longer permit the following modifications:

- Order postings on the terminal
- Order posting in the MOC (e.g. using the "order overview" function)
- Modifications to the event maintenance

Furthermore, the graphic planning board of the HYDRA Shop Floor Scheduling module (HLS) does no longer show the machine/ workplace .

Company

This field differentiates the individual machines/ workplaces. It also functions as report/evaluation option in the system.

Group

Assignment of the workplace/ machine to a logical group. In the context of the planning this has to do with a capacity group in which the primary capacities are summarized.

If you create a new workplace, it is automatically assigned to a group of the same name (menu BDE: Master data > Workplaces/machines > Groups), which is defined as a capacity group. If there is no capacity group yet, then the system automatically creates a capacity group and assigns the workplace automatically.

Category

Enter the category of the machine. By means of this, you can enable a validation check according to the BDE configuration: *Master data > Order configuration > Order types, validation tab, Option "Check specification in backlog of orders"* (value category).

Year model

Enter a valid year model. During entry, times to be posted are compared with this shift model. If no planned year model is stored in the HLS tab, the shift model entered here is also used in the HYDRA Shop Floor Scheduling module (HLS).

Standard rate machine

Here enter the arithmetical standard rate of machines for calculations. The HYDRA Shop Floor Scheduling module (HLS) uses this value for some (evaluated) KPIs.

Standard labor rate

Here enter the arithmetical standard labor rate for calculations. The HYDRA Shop Floor Scheduling module (HLS) uses this value for the KPI "Evaluated labor utilization".

Performance level

Enter the performance level of the workplace/ machine in percent in this field. This value is integrated in the HYDRA Shop Floor Scheduling and in the evaluation of material requirements when calculating the remaining run time.

Incentive wages indicator

Defines the type of calculating incentive wages. Mostly, this option is used together with the incentive wages based on formulas for customer-specific configurations. In addition, use the "incentive wage indicator" as selection criterion for the *identification of wage types* in the incentive wage determination.

Leave this field empty, if you do not use the incentive wages module.

The *incentive wages indicator G=group calculation* has a special meaning. If the workplace/machine has this option, you have to assign a premium group every time an order is logged on. You can do this either via

- the "assignment of premium groups" option of the incentive wages module or, optionally, via
- an additional field in the terminal dialog for the logon of orders. If no assignment is available, the system rejects the logon of the order by issuing a validation error.

Therefore, you may only assign the incentive wage indicator G = Group calculation, if the group premium conditions are met in the incentive wages determination, as otherwise orders can no longer be logged on!

The meaning of the other incentive wages indicators is specified according to the customer's requirements while customizing the system.

File

You can assign a graphic to each machine/workplace. Among other uses, the graphic is displayed in the workplace overview or in the AIP. The following image formats are supported: jpg, gif, tif, bmp, ico, emf, wmf.



In the path configuration, configure the path using the PATH identification "MOCWPIMG"; the length of the file name for graphics files is limited to 12 places (8.3 notation). Note for Unix installations: use lower case letters only for file names.

Maximum capacity (KG)

If a machine is configured as melting aggregate, define the maximum capacity in KG here.

Accuracy class, unit, etc.

Information fields in order to describe the accuracy. These fields are only available if Test Equipment Management (PMV-PPK or PMV-SVP) is licensed and the right "mdresgenh" is assigned.

Entry

Display 3rd list

Use the options described here to show/enable a third list in the basic screen of a Windows terminal (CTWIN / AIP). Switch between the respective lists on the terminal depending on the options set. The following settings are possible. Please note that the contents displayed in the lists depend on the module in use:

- Input material (MPL): shows logged on input materials/ batches.
- Resources (WRM): shows logged on resources and tools.
- Staff (BDE): shows logged on staff.
- Output material (MPL): shows produced output batches.

Show material/ PRT list when OP is logged on

This option is only relevant in connection with the WRM module and the resources logged on to the Windows terminals (CTWIN / AIP).

If this option is set and you log on an OP, a specific login dialog opens at the AIP terminal. This dialog includes a list of components/production resources and tools. This list shows resources that meet at least one of the following requirements:

- the option "posting to terminal" is set in the resource type;
- the option "log on with OP" is set to "explicit logon" for the resource.- the resource is a so-called "required resource" (option of the resource).

Please note: As long as the workplace is relevant for MPL, the list also shows material components.

Sequencing list

This option defines which operations are displayed in the sequencing list on the terminal. The following settings are available:

- S Basic setting. The system takes the value from the option of the same name in the *HYDRA basic settings*.

- M Pool of workplaces. The sequencing list of the terminal only shows the operations planned for the workplace.
- G Pool of workplaces and groups. The sequencing list of the terminal either shows operations that are:
- planned for the current workplace or
 - for another workplace of the group or
 - that are still located in the pool of groups.
- K Pool of workplaces and categories. The sequencing list of the terminal only shows those operations that are planned for workplaces of the category.
- H Group control. The sequencing list of the terminal shows the operations that are either planned for the current workplace or for another workplace in the group.

Number of OPs in sequencing list

Here, define the maximum number of operations that are to be displayed in the sequencing list on the terminal. Enter 0 if you want to show all operations.

Compulsory sequence

Use this option to specify whether or not logging the OPs on in the planned sequence is compulsory. The following parameters are permitted:

- N Disabled
J Enabled/active

If the parameter is "enabled" when you log on an OP, the system checks as to whether the order backlog for this machine/ workplace includes an OP that is planned for the same time or previous to this OP in the sequence, but was not yet started (i.e. status = V/prepared). If yes, the system rejects the logon of this OP.

Please note: If you use the order sequencing option (menu ADE: Planning > Order sequencing) for planning in HYDRA and configure the sequencing list with any other option than "M" (pool of workplaces) and enable the compulsory sequence, this might lead to a combination that does not make sense.

Please note for the sequencing list:

- If the sequencing list includes operations that are in the status "interrupted", you can log on these OPs at any time, irrespective of the specified compulsory sequence.

Dialog control

To meet this requirement, define a dialog control that deviates from the standard behavior for the workplace in the dynamic dialog configuration of the Windows terminal (CTWIN / AIP). Then refer to the dialog control in the dialog.

Use this configuration only as part of customizing the HYDRA system. Otherwise it has no significance.

Logon of several OPs

Select this option, if several different operations should be processed on the machine. Otherwise, the system only allows one operation to be logged on to the machine.

Possible values:

J Log on as many OPs as required at the same time.

Please note: The system allows a maximum of 20 operations to be logged on simultaneously to a machine to which a terminal with operation mode MDE is assigned. If more than 20 operations must be logged on at the same time, MPDV must review the conditions in order to remove the limitation. If MPDV agrees to lift the limitation, you can do so, otherwise search for alternative solutions. MPDV analyzes the conditions as part of a service.

N Log on only 1 OP.

1...9 Log on a maximum of n OPs.

Posting

Quantity posting to staff

Use this function to post the quantity of order interruptions/ logoffs to the person who is logged on for the longest period.

Detailed information about quantity posting to staff can be found [here](#).

Posting on OPs that are not logged on

Use this option if you want to post quantities (interrupted OP, finished OP, reporting partial quantities) even for operations that are not logged on.



If you record quantities for an operation that is not logged on, the system posts these quantities onto the operation in the BDE module. The MDE module does not post the quantities.

If you want to use this function with the AIP terminal, the BDE posting dialogs that are installed by default require the following:

- use the simplified BDE posting dialogs (the so-called "[simple dialogs](#)") or
- customize the dialogs.

Then you will be able to enter an operation that is not logged on.

Posting the machine time in connection with operations logged on simultaneously

Use this option to post the machine time for OPs that are logged on simultaneously as a proportion on the operations and staff:

J Posts the machine time proportionately on OP and person according to the number of OPs

- N No proportionate posting. If the option is not set, every operation receives the complete machine time.
- V According to the default quantity of the OPs. Make sure that the default quantity (target quantity in primary quantity unit) of the operation > 0.
- Z According to the standard time of the OPs (available starting with ADE 7.3). Here, ensure that the standard time (processing time) in the operation > 0.

Please note:

This option is also evaluated for group workplaces; in general, do not use this option for group workplaces.

Automatic logoff of personnel when shift ends

This option is only relevant, if you set an "X" for the option of the same name in the order type.

Use this option to configure the personnel-related data collection at MDE workplaces. Because fully automatic shift ends are generated by the terminals when using the HYDRA MDE, you can configure here if

- the staff logged on to the workplace should be logged off automatically at the end of the shift or
- if they should remain logged on.

- J Always log off staff when the shift ends
- N Always save staff when the shift ends except for manual logoff
- X Evaluate the person's settings. The system searches for the corresponding setting of the person .

Automatic OP posting when shift ends

This identifier is only relevant, if you set an "X" for the option of the same name in the order type.

- J Interrupt and log on again at beginning of shift
- N Interrupt

MPL

Further information about the HYDRA module MPL can be found in the corresponding MPL documentation.

Batch management

Activates the entry of the batch number for this machine within the posting dialogs on the terminal.

Possible values are:

- N No batch processing
- L Batch tracing (input/ output batches) in the context of HYDRA MPL/MPL-RF
- D Throughput batch processing in the context of HYDRA MPL/MPL-RF
- J BDE batch management

The following functions are only available in connection with the Material and production logistics module and are supported only on the Windows-based terminals (CTWIN / AIP).

Preceding material buffer

Irrelevant.

→Subsequent material buffer

If you specify a material buffer in this field, the field *Target buffer* in each of the entry dialogs (e.g. output batch change, log off operation) is automatically populated with this value.

If you do not enter a material buffer in the input dialog (e.g. delete from the input field), the system automatically posts the output batch to the material buffer specified in the "subsequent material buffer" field.

Automatic generation of batch number

If you set this option, the system automatically generates a batch number for the output batch to be produced. Otherwise, the system expects you to enter the batch number for the new output batch to be produced, when you log on an operation or change the output batch.

Please note: If, in the field *Batch management* you set the option D (= *Throughput batch recording*), the system automatically sets the value for the *Automatic generation of batch number* to "J". In this case, you cannot enter the batch number manually.

Consumption balance

When you log off an OP, the system opens an additional dialog (V_BLZ) displaying the material components and their consumption quantities in relation to the OP that is currently logged on. In this dialog, you can also log off input batches that are still running. This option is only activated, once you have enabled the consumption balance for the material type of the output material.

Generate transport order for output batches

This option creates a transport order relating to batches for a generated output batch. The transport starts from the material buffer where the output batch is included. The configurations of the material type override the corresponding options of the resource configuration.

Generate transport order for input material

This option creates an article-related transport order relating to a material component, when you plan an operation for a machine via the Shop Floor Scheduling module. Transport starts from the output material buffer of the preceding operation. The configurations of the material type override the corresponding options of the resource configuration.

HLS

Find further information about the HYDRA module HLS in the corresponding HLS documentation.

Planning function

This identifier specifies whether a workplace or a machine will be displayed and if so, in which MOC planning function.

- | | |
|---|--|
| P | Planning in the graphic planning board of the HYDRA Shop Floor Scheduling (HLS) or in the graphic order sequencing (GAV), i.e. plan the workplace via the HLS or the graphic order sequencing. Therefore, the HLS module or the graphic order sequencing also show this workplace but this workplace is not visible in the tabular order sequencing (AVG). |
| | <p>Please note: If the HLS module or the graphic order sequencing actually show the workplace also depends on additional settings:</p> <ul style="list-style-type: none"> - the workplace must be assigned to a group identified as a "capacity group" - you must have the responsibility area authorization for this workplace - you must have selected the corresponding planning profile |
| H | Only relevant, if you use the HYDRA Shop Floor Scheduling (HLS).
Like P. |
| T | Reserved. |
| A | Planning in the order sequencing table (AVG), i.e. plan the workplace via the AVG module. |
| N | No planning; the order sequencing table (AVG), the graphic order sequencing and the HLS module do not show the workplace. |

Planned year model

Enter a special year model used only for planning in the HYDRA Shop Floor Scheduling (HLS) module. This year model does not affect data collection and posting in the ADE/MDE module. If you do not define a planned year model, then the system uses the BDE year model (*Master data* tab) for the planning.

Availability

Define the available capacity of a workplace/machine. The default value for the available capacity is 1000 [per mill].

In the HYDRA Shop Floor Scheduling, the capacity check and automatic assignment assume that each operation has a capacity requirement of 1000 [per mill], i.e. exactly one operation can run on the workplace/machine at a time. In case of a manual multiple assignment, a dialog informs you about the double assignment. If you use the automatic assignment, multiple assignments are generally not feasible.

Use this setting to extend the availability of the workplace such that a multiple assignment is permitted. If the workplace capacity allows, for example, processing of two operations at the same time, set the available capacity to 2000 [per mill] in this field.

If nothing is entered in this field or if you enter the value 0, the system interprets this as the default value of 1000 [per mill].

This functions requires a corresponding license.

Check personnel availability

Choose from the following options:

- Check if at least one person is planned
- Check personnel availability
- Check personnel availability and qualification

If you select one of the above-mentioned options, the system checks if staff is planned in the application [Workplace assignment](#) at the time when you [plan](#) operations in the [graphic planning](#). Find further information on personnel capacities in Graphic Planning [here](#).



This option is only available if the modification [hls_pep_qualification](#) is enabled.

"Quantities" tab

This tab is only available if you select a resource of the type "MNR".

Conversion factors for base quantity

Enter the quantities in various quantity types and accounts at the machine or workplace. In general, the system supports the following **quantity accounts**:

Yield

Scrap

Rework (Windows terminal CTWIN/AIP only)

Open quantity (problem quantity; Windows terminal CTWIN/AIP only)

Each quantity account supports the following **quantity types**:

Primary quantity

Secondary quantity (Windows terminal CTWIN/AIP only)

Tertiary quantity (Windows terminal CTWIN/AIP only)

Basic quantity (Windows terminal CTWIN/AIP only)

The system design specifies the use of several quantity types or accounts. For example: If you want to enter the rework quantity manually, a corresponding input field must be configured in the input dialog (customization).

Use the quantity type "primary quantity" if you want to collect quantities automatically.

Quantity units and conversion factors for base quantity

Define a quantity unit for each quantity type. You can enter data directly (i.e. manually) in the alternative quantity accounts. If this is the case, the system does not convert quantities automatically.

If you do not enter data manually in the alternative quantity accounts, the server converts the quantities into the alternative accounts based on:

- the conversion factors or
- the units that are configured in the MOC machine master data:

Basis for HYDRA-MDE quantity conversion

Define the basis for the quantity conversion.

- A Use the conversion factors of the OP that is logged on. If no operation is logged on, the system uses the quantity conversion stated in the machine/workplace configuration.
- M Use conversion factors from the workplace configuration for the quantity conversion.

Units and conversion factors for base quantity (P)

Quantity unit (P)

Indicate the quantity unit you want to use for data collection at this machine/ workplace. If you collect quantities automatically, these quantities are generally primary quantities.

If you want to convert quantities automatically into another quantity type, indicate the conversion factors for the base quantity here.

Units and conversion factors for base quantity (S)

Quantity unit (S)

Indicate the secondary quantity unit you want to use for posting the quantities to the workplace/ machine. If you want to convert quantities automatically, indicate the conversion factors for the base quantity here.

Units and conversion factors for base quantity (T)

Quantity unit (T)

Indicate the tertiary quantity unit you want to use for posting quantities to the workplace/ machine. If you want to convert quantities automatically, indicate the conversion factors for the base quantity here.

Units and conversion factors for base quantity

Quantity unit (B)

Indicate the base quantity unit you want to use for posting quantities to the workplace/ machine.

Manual entry of quantities, yield

Manual entry of yield

Set this option, if you want

- to collect quantities manually;
- to set off the quantities against another account;
- to post the manual quantities as cycles.

On Windows terminals this option does not affect the quantity fields displayed in the input dialogs. Change these quantity fields via dialog configurations (terminal configuration or customizing of the dynamic dialogs).

Allocation of yield

Requirement: Set the option "Manual entry".

Use this option to offset manually entered quantities against other quantity accounts. In this case, the entered quantity is deducted from the specified account.

Keep in mind that the result of relevant quantity compensation may also be a negative value.

Please note

Do NOT set this option for DOS terminals, if yield is offset against scrap or scrap is offset against yield in the counter configuration.

Posting of yield as cycles

Requirement: Set the option "Manual entry".

If this option is set, the system also posts manually entered quantities as cycles. Note here that the entered quantity is posted 1:1 as a cycle (partitioning is not considered).

Manual entry of quantities, scrap

Manual entry of scrap

Set this option, if you want

- to collect quantities manually;
- to set off the quantities against another account;
- to post the manual quantities as cycles.

On Windows terminals this option does not affect the quantity fields displayed in the input dialogs. Change these quantity fields via dialog configurations (terminal configuration or customizing of the dynamic dialogs).

Allocation of scrap

Requirement: Set the option "Manual entry".

Use this option to offset manually entered quantities against other quantity accounts. In this case, the entered quantity is deducted from the specified account.

Keep in mind that the result of relevant quantity compensation may also be a negative value.

Please note

Do NOT set this option for DOS terminals, if yield is offset against scrap or scrap is offset against yield in the counter configuration.

Posting of scrap as cycles

Requirement: Set the option "Manual entry".

If this option is set, the system also posts manually entered quantities as cycles. Note here that the entered quantity is posted 1:1 as a cycle (partitioning is not considered).

Manual entry of quantities, rework

Manual entry of rework quantity

Set this option, if you want

- to collect quantities manually;
- to set off the quantities against another account;
- to post the manual quantities as cycles.

On Windows terminals this option does not affect the quantity fields displayed in the input dialogs. Change these quantity fields via dialog configurations (terminal configuration or customizing of the dynamic dialogs).

Allocation of rework

Requirement: Set the option "Manual entry".

Use this option to offset manually entered quantities against other quantity accounts. In this case, the entered quantity is deducted from the specified account.

Keep in mind that the result of relevant quantity compensation may also be a negative value.

Please note

Do NOT set this option for DOS terminals, if yield is offset against scrap or scrap is offset against yield in the counter configuration.

Posting of the rework quantity as cycles

Requirement: Set the option "Manual entry".

If this option is set, the system also posts manually entered quantities as cycles. Note here that the entered quantity is posted 1:1 as a cycle (partitioning is not considered).

Manual entry of quantities, open quantity

Manual entry of open quantity

Set this option, if you want

- to collect quantities manually;
- to set off the quantities against another account;
- to post the manual quantities as cycles.

On Windows terminals this option does not affect the quantity fields displayed in the input dialogs. Change these quantity fields via dialog configurations (terminal configuration or customizing of the dynamic dialogs).

Allocation of open quantity

Requirement: Set the option "Manual entry".

Use this option to offset manually entered quantities against other quantity accounts. In this case, the entered quantity is deducted from the specified account.

Keep in mind that the result of relevant quantity compensation may also be a negative value.

Please note

Do NOT set this option for DOS terminals, if yield is offset against scrap or scrap is offset against yield in the counter configuration.

Posting of open quantity as cycles

Requirement: Set the option "Manual entry".

If this option is set, the system also posts manually entered quantities as cycles. Note here that the entered quantity is posted 1:1 as a cycle (partitioning is not considered).

"MDE configuration" tab

This tab is only available if you select a resource of the type "MNR".

Monitoring

Monitoring type

Choose from the following monitoring types:

Monitoring by operating signal

No monitoring

Cyclic monitoring

If you select **cyclic** or **operating signal** monitoring, you can only enter a malfunction if the terminal prompts you to do so ("Assign malfunction"). If you do **not** use **automatic monitoring**, you can enter a new machine status at any time.

If you use the **cyclic** monitoring option, the machine automatically switches to the "production" status when counting pulses occur. If you select the "operating signal" option, the machine automatically switches to the status "production" as soon as the operating signal is set. If you do not use the "automatic monitoring" option, you must assign the "Production" status manually.

→Entry of malfunction reason required with specified delay time in [s]

(Only with terminal type CT 73x, CT 83x, not with master terminal/DS-100)

If the system identifies a downtime without a reason, the terminal opens the input dialog "Change machine status" after the specified delay time. If the terminal goes back into production, the window still remains open.

If you now enter a machine status (during production), this input activates a reposting event that reposts the most recently recorded status from "General disturbance" to the newly entered status. If reposting is successful, the window closes; otherwise, it remains open.

However, if the system identifies a downtime once again (with or without a reason), you can no longer repost the previously noted status. The window closes automatically.

If the system identifies another downtime without a reason and the delay time has expired, then the input window opens as described above.

If the system identifies a downtime without a reason and the machine switches into production before the delay time expires, then the terminal does not automatically prompt you to enter a malfunction reason.

Important note:

This reposting only affects the HYDRA Machine Data Collection. The system does not correct the resource performance accounts of the currently running OP online!

Please note for data maintenance:

The tabular event maintenance of the MOC shows all changed machine statuses. However, you cannot edit the reposting event as it is locked. In order to perform recalculations correctly with respect to orders and machines, change the original event with the status "NOT ASSIGNED" to the correct status. The reposting event does not affect recalculation!

Minimum malfunction time

If you select the **operating signal** monitoring type, define the duration in seconds that a malfunction must last until it is identified and posted as a malfunction.

Minimum cycle time

If you select the **cyclic** monitoring, specify a minimum cycle time in seconds in this field.

The terminal uses this minimum cycle time and the target cycle stored in the (logged in) operation and compensated with the cycle extension in order to calculate the maximum value. The terminal uses this value as the default cycle time.

If both the minimum cycle time and the target cycle stored in the operation are 0, the cycle time specification is set to 60000 seconds [per 1000 machine cycles].

Cycle extension

If you select the **cyclic** monitoring option, enter the percentage for extending the target cycle time in a range from 0 to 5000.

The system sets off the target cycle stored in the (logged in) operation against this percentage. In this way, a value less than 100 indicates a shortened cycle; a value greater than 100 indicates an extended cycle.

Number of target cycles

If you select the "**cyclic** monitoring" option, enter the number of cycles (0 to a maximum of 9) after which the terminal automatically switches from a status unequal to "production" into the "production" status within the cycle time (requirement: this status unequal to production is not locked for the "production" status).

Some production processes provide machine cycles during the set up phase. Set a value greater than 0 in order to prevent the current machine status from changing immediately. Please note: The quantities you collect until the machine switches to the "production" status are neither posted as yield nor scrap.

Cycles to be evaluated

Reserved. Enter 0 in this field.

Administration**Posting during production lock**

Use this setting to specify how to post the counting pulses that are collected while the status "production" is suspended. This configuration takes effect with all counters configured as "Yield".

Posting as scrap If this option is configured for the counter, the system offsets the counting pulses against the partitioning/ pulse factor and posts these pulses as scrap. Even if you defined another quantity account for offsetting, this one will not be used.

Posting as yield parts the system posts the counting pulses as yield

No posting the system does not post the quantities while the "production" status is suspended.

→→→Pulse factor specific to machines

Use the pulse factor, for example, if you want to collect lengths (e.g. using a wheel).

Set the value to 0 for machines where a discrete or integral number of quantities (e.g. pieces) is collected per pulse. In this case, the pulse factor is not evaluated. That means, the number of cycles posted corresponds to the actual pulses transferred via the MSS (machine interface).

The MSS (machine interface) records the signals transferred from the machine (counting pulses). According to the configured number of pulses, the system calculates and posts the quantities as follows:

Quantity for the machine = pulse * partitioning for the machine/ pulse factor for the machine

Quantity for the operation = pulse * partitioning for the operation/ pulse factor for the operation

Please note: The pulse factor will be evaluated as a fraction. In this way, in the quantity calculation the pulse is included as a denominator while the partitioning is considered a numerator.

The system interprets pulses that occur during a malfunction or a production lock (configuration of *Posting during prod. lock* > scrap) as scrap. Also use the above-mentioned formula to calculate the scrap quantities.

Partitioning specific to machines

Enter the partitioning specific to the machine in this field. Multiply the machine-specific partitioning by the partitioning stored with the operation in order to integrate the machine-specific partitioning into the quantity calculation. Enter the value 1 in this field, if you do not want this to happen.

Extended weekend automatic

If you select this option and if configured accordingly, the system assigns the status that was available before status 999 was activated when the shift starts.

Please note:

To use this option, the workplace must already be assigned to a terminal.

Find detailed information about the automatic activation of status 999 in the document [Day types](#).

Waiting period, short-term disturbance

To improve the overview, e.g. in the machine history, configure a short-term malfunction status for each machine/ workplace. Use this status as a "repository" for unconfirmed statuses, which only existed for a specific (short) period.

If the terminal automatically identifies a downtime and the machine automatically goes back into production, the system checks if this disturbance is shorter than the time period configured here for short-term malfunctions.

If so, then the system assigns the status that is configured for "short-term disturbances" to this automatically identified malfunction.

Inputs/ outputs

Machine lock/ Target quantity reached/ Machine downtime/ Free I/O

Enter the logical output where a digital signal should occur when the corresponding status is available.

Machine lock output the system sets this output, if you enabled the option "machine lock output" in the current machine status.

Target quantity reached output the system sets this output, if the collected yield has reached the target quantity of the OP.

Machine downtime output the system sets this output, if the machine is in a status unequal to **Production**. When changing into production, the system immediately sets the output to 0.

Free I/O Free input/ output for customizations.

Use these statuses for connecting a monitoring light or a horn, for example.

Enter the corresponding number in one of the fields in order to assign an output and to specify which relay is interconnected by the terminal when the predefined status occurs. Enter "0" to prevent any action. Note that you cannot assign an output on a terminal more than once.

Please note

- Specify the statuses that trigger the activation of the machine lock as part of the **Status assignment**.
- Generally, enter the value "1" in the input field, when the machine lock is activated via the available relay output of a DS 100. In this case, the system sets the machine lock if
 - a correspondingly configured status occurs and
 - in the status not assigned.

Change output batch

Customer-specific assignment of an input with an automatic output batch change (MPL). By default, enter 0 in this field.

PDE (Process Data Collection)

Collect process data

This parameter specifies if the system collects process data for this machine. If this parameter is not set at a machine, you cannot collect process data for this machine.

External connection

External connection

If this machine is assigned to a master terminal, you can choose from the following connection options:

Arburg control system	Arburg connection (only available if HYD-ALS is licensed)
Engel interfacing	connection of Engel machines
No external device	No connection of an external device
DS100	DS100 connection
MT3	MT3 connection
PDE	Process data collection

If you activate a DS100 or MT3 connection, you can select the field "device address". If you activate the option "Engel interfacing", you can select the field "serial number". If you activate the option "Arburg control system", you can select the field "class".

Note regarding the combination of connections on a master terminal:

"DS 100" and "No external device": allowed

"MT 3" and "No external device": allowed

"MT3" and "DS 100" not allowed!

Serial number (Engel interfacing)

Enter the serial number of the connected Engel machine. Set the option "EMS machine interface" in the HYDRA basic parameter settings if you want to use Engel machines.

Device address

You can select this field, if you activated a DS100 or MT3 connection. Enter the device address of the sub-bus participant.

"Resource configuration" tab

This tab is only available, if you select a resource of any **other** type **than** "MNR".

Resource master data

Type

Identifier regarding the type of resource:

Resource: A resource can be uniquely identified, i.e. the resource is actually present. It always has the number 1.

Anonymous resource: An anonymous resource cannot be uniquely identified. If the identifier is set, then you can change the value in the field *Number* from 1 to another positive integer value. You cannot post data onto anonymous resources because anonymous resources do not relate to one specific resource. Please note the information in the chapter *Anonymous resources*.

Required resource: A required resource is a substitute for one or more actual resources that can be identified. Specify in the configuration *WRM: Master data > Required resources* which resources are represented by a required resource. The number results from the number of actual resources assigned to the required resource.

Please note: If this field is empty, the resource is implicitly an ("actual") resource.

Equal type

Reserved for future use.

Version

Revision number; store here the program version for resources of the type DNC.

Number

You can only edit this field, if it contains an anonymous resource and the identifier *Anonymous resource* is set (see above). A value > 1 indicates how many of these resources are available.

This field is automatically calculated for required resources.

Resource family

Assign a resource family. If you change the resource family subsequently, an information dialog appears as a warning because user fields can possibly be assigned with the resource family.

Target utilization

Cycles

The target cycles serve as additional information regarding how long the resource is to be used.

Runtime

The target runtime serves as additional information regarding how long the resource is to be used.

Input unit

Input unit

Absolute value limit (EMG 8.1, function authorization: resablim)

Enter the absolute value limit of the (meter) resource. The energy monitor shows this limit value in addition to the current meter reading. Use the Escalation Management to generate an escalation message, if the counter value of the resource exceeds the specified absolute value limit. This field is only visible, if you have the function authorization "resablim".

Configuration

Target cycle

Target duration in seconds for 1000 machine cycles if this tool is used.

Please note: The target cycle stored in the OP is relevant for the planning in the HLS module and for the machine data collection at the terminal.

Original partitioning

Partitioning of the tool (= number of cavities) when using this tool.

Current partitioning

Current partitioning of the tool. This partitioning can deviate from the original partitioning, e.g. if the original quantity can no longer be produced with one cycle due to a tool defect.

Always use the current partitioning to post cycles to the tool.

Please note: The partitioning stored in the OP is relevant for the planning in the HLS module and for the machine data collection at the terminal.

Partitioning due to cavities

If you set the option "partitioning due to cavities", the system (re-)calculates the fields "current partitioning" and "original partitioning" using the assignments in cavity management. Then, you can no longer change the fields manually.

Log on with OP

Use this identifier to control whether or not the resource is logged on with the OP, if it is included as a component in the list of production resources and tools of the operation. Possible values:

None: The resource is not logged on.

Implicit: The system automatically (implicitly) logs on the resource that is assigned to the operation as a production resource and tool; you can neither log on the resource explicitly nor change the logon.

Explicit: You can explicitly log on the resource that is assigned to the operation as a production resource and tool or you can log on another resource instead. If you do not log on the resource explicitly or if you do not log on another resource explicitly, the system implicitly (automatically) logs on the current resource; in this way, the current resource serves as a "default".

Please note:

If you log on another resource explicitly, this resource will be logged on for the resource that has the same **resource type** in the operation's list of production resources and tools. For this reason, you can only log on those resources explicitly that are included as a requirement in the operation's list of production resources and tools. In this way, you cannot log on a resource that is not included as a requirement in the list of production resources and tools.

In general, you should not enable this option for the resource type DNC. The DNC module handles this differently (NC programs are logged on separately).

The system also logs on resources that are defined in the BOM of the machine.

Parallel logon/ planning possible

You can log on/plan the tool simultaneously.

Please note: You can only log on a resource multiple times to only one machine. Consequently, the option "Parallel logon possible" refers to several different OPs on one machine.

In this case, the system posts data proportionally as follows:

- Post quantities proportionally.
- Post times 100% for each resource. This means that the system posts double the time to the resource, if the resource is logged on twice.

Post to resource

Indicates whether or not the quantities and times are to be posted to the resource. Due to a high degree of complexity, you should only assign this identifier to those resources that you actually want to evaluate.

Planning

Setup time

Duration in hours for setting up the tool.

Please note: The setup time stored in the OP is relevant for the planning in the HLS module.

Retooling time

Duration in hours for removing the tool.

Please note: The retooling time stored in the OP is relevant for the planning in the HLS module.

Assignment

Not used. The system uses the configuration option of the same name stored in the resource type in order to integrate the resource allocation in the HYDRA Shop Floor Scheduling.

Evaluation

Consider in evaluations

Reserved for future use.

File

File exists

Shows whether or not the file is stored in the specified path. A cyclic process checks the files and sets the options subject to whether or not the file is available.

File name

File name; without file extension for DNC. The system adds the file extension based on the configuration in the resource type. The defined paths specify the storage location.

Comparison resources

Enter two comparison resources for energy consumption resources. They will then be shown in comparative evaluations/reports, e.g. the energy monitor.

Resource 1

Resource number of the resources to be compared.

Resource type 1

Resource type of the resources to be compared.

Resource 2

Resource number of the resources to be compared.

Resource type 2

Resource type of the resources to be compared.

Accuracy

Enter more detailed information on measuring accuracy and measuring range for test equipment resources.

"User fields" tab

Use user fields to store further customer-specific information to the MES besides the available fields in the MOC standard. The "user fields" tab includes eight sub-index tabs, which each have eight additional user fields. The so-called user field key determines which user fields are involved and which meaning they have.

The workplace and resource configuration provides data of two basic object types. You can also edit this data in the workplace and resource configuration: on the one hand these are machines and workplaces and on the other these are the resources. Machines and workplaces are also "resources". But resources are not automatically machines and workplaces.

Object type

The system configures the user fields of machines/workplaces in relation to the object type "MNR". The system stores data contents to the machines/workplaces table and the resources table of the database in order to ensure data consistency.

Configure user fields for resources in relation to the object type matching the resource type of the resource (example: create resources of the type "PAC" in relation to the object type "PAC"). The system stores data contents to the resources table of the database.

User field key

Every user field key describes a combination of user fields. The management of the user field key (and therefore the purpose of the fields) varies from one object to the next.

User fields

The following user fields are available after configuration:

Field data type	Number of fields
Date	6
Numeric, time, duration	16
Decimal value	6
Text field, length 1	16
Text field, length 10	6
Text field, length 20	14
Text field, length 40	2

Each page shows a maximum of 8 fields.

User field keys are not defined by default in the system. Configure the system accordingly to support this kind of user fields.



As the table shows resources of different types, use the user field key "SYSTEM" of the object "RES" to identify the column headings for the user fields. .

"Comment" tab

Store additional resource comments in the "comment" tab.

"Resource attributes" tab

Shows additional resource attributes via the user field definitions of the resource family. Use the "resource attributes" button for editing.

"Resource list" tab

Shows the resource list for the selected resource. Click the "resource list" button to go directly to the BOM application for editing purposes.

"DNC versions" tab (available as of DNC 8.2)

Shows the available versions of a DNC resource including a flag indicating the currently applicable version. HYDRA provides this valid version for machine downloads.

Toolbar

"General" tab



Insert

Opens the dialog for adding a resource. This dialog provides the fields that match the selected resource type.



Copy

Function authorization: mdres.copy

Opens the dialog for copying an existing resource. Subject to the resource and corresponding resource type selected before starting the "copy" function, the function differentiates the following:

- Copy function for resources of the **Type = MNR** (workplaces, machines)
- Copy function for resources that do **not** have the **Type = MNR** (resource type != MNR)

Copy function for resources of the Type = MNR (workplaces, machines)

From: resource type, resource, short name, name

- Resource type (fixed "MNR")
- Workplace/ machine number
- Short name
- Name

of the workplace you want to copy. You cannot change these values. They derive from the selected data record.

To: resource type, resource, short name, name

- Resource type (corresponds to the resource type of the workplace you want to copy; cannot be changed).
- Workplace/ machine number
- Short name
- Name

of the target workplace.

Copy machine status

Function authorization: mdmst.copy

If you set this option, the system automatically creates and transfers all [workplace/ machine statuses](#) of the workplace you want to copy to the new workplace.

Copy counter configuration

Function authorization: mdctr.copy

If you set this option, the system automatically creates and transfers all [Counters/meters](#) of the workplace you want to copy to the new workplace.

Note that the counter numbers of the new workplace are identical with the counter numbers of the workplace you copied. If necessary, you have to adjust these counter numbers.

Copy reasons

Function authorization: mdreas.copy

If you set this option, the system automatically creates and transfers all [reasons](#) of the workplace you want to copy to the new workplace.

Copy function for resources of any other type than MNR (resource type != MNR)

The copy function for all resources that do not have the type MNR opens the "insert" dialog and takes over the details from the previously selected resource. But you can edit and change all fields.

**Edit**

Opens the dialog for editing a resource and provides the tabs and fields of the corresponding resource type.

**Delete**

Deletes one or several selected resources.

"Resource" tab**Configuration – resource status**

Opens the application "resource status" to define statuses for all resources that do not have the type MNR.

**File - show file**

Shows the file - only available for document resources configured as file-based resource without DNC processing in the resource type and if the corresponding license and function authorizations are available.

**Go to - resource list**

Opens the "resource list" application. The selected resource is entered as default value for the superordinate resource.

**Go to – required resources**

Opens the "required resources" application. The selected resource is entered as default value for the required resource.

**Go to – cavity assignment**

Opens the "cavity assignment" application. The selected resource is entered as default value.

**Go to - resource attributes**

Opens the application "resource attributes". The selected resource is entered as default value.

**Functions – measures**

Opens the "measures" application.



Functions – status change

Opens the dialog for changing the resource status. The checkbox "including subordinate resources" is not relevant and reserved for future upgrades.



Functions – release of resource

Opens the dialog for releasing a resource. The checkbox "including subordinate resources" is not relevant and reserved for future upgrades.



Functions – stock transfer

Opens the dialog for transferring/relocating a resource

"Workplace" tab



Configuration – status assignment

Opens the application "status assignment". The system enters the selected resource in the corresponding field.



Configuration – counter configuration

Opens the application "counter configuration". The system enters the selected resource in the corresponding field.



Configuration – terminal assignment

Opens the application "terminal assignment". The system enters the selected resource in the corresponding field.



Entry – reasons

Opens the application "reasons". The system enters the selected resource in the corresponding field.



Entry – Operator positions

Opens the application "operator positions". The system enters the selected resource in the corresponding field.



Entry – premium indicator

Opens the application "premium indicator". The system enters the selected resource in the corresponding field.



Groups - groups

Opens the application "groups". The system enters the group of the selected resource.



Groups – group assignment

Opens the application "group assignment". The system enters the selected resource in the corresponding field.



Miscellaneous – cycle parameter

Opens the application "cycle parameter". The system enters the selected resource in the corresponding field.



Miscellaneous - workforce requirements of workplaces

Opens the application "workforce requirements of workplaces". The system enters the selected resource in the corresponding field.

"DNC" tab

The tab is only available, if you select a DNC resource. These are resources configured as resources with DNC processing in the resource type.



Configuration – resource status

Opens the "resource status" application.



Configuration - assignment of DNC family to machine

Opens the application "assignment of DNC family to machine".



File - comparison editor

Opens the comparison editor for the selected resource or resources. See below for further information.



File - export

Exports the file entered for the resource. Enter the target file via the file explorer.



File - import

Imports the file entered for the resource. Select the source file via the file explorer.



File - viewer

Opens the file entered for the resource for viewing using the defined viewer program.



File - editor

Opens the file entered for the resource for editing using the defined editing program.



Go to - resource attributes

Opens the application "resource attributes". The selected resource is entered as default value.



Go to - resource list

Opens the "resource list" application. The selected resource is entered as default value for the superordinate resource.



Functions – status change

Opens the dialog for changing the resource status.



Functions – release of resource

Opens the dialog for releasing a resource.

How to use the comparison editor

The comparison editor compares the files attached to the DNC resources. You can choose from two options:

Select one resource:

The editor shows the released resource and the optimized version of the resource for comparison. You can change the file entered on the right-hand side of the editor. Once you have made the changes, the comparison editor transfers these changes to the system, like the simple editor. You can only use this option for DNC types with the file processing type "optimized".

Select two resources:

If you select two resources before you open the editor, the editor compares the two selected resources. You can choose the file type. You can change the file entered on the right-hand side of the editor. Once you have made the changes, the comparison editor transfers these changes to the system, like the simple editor.

Start the functions of the comparison editor by clicking on the relevant buttons or via the context menu (right clicking):

- Reject: Reject the detected difference (on the right). Accept the value from the left file. The editor does no longer highlight the difference.
- Keep: Accept the detected difference (on the right). The editor does no longer highlight the difference.
- Next difference: Go to the next difference.
- Insert: Inserts a row at the current position.
- You can always change the contents of a row by clicking the row and inputting values. Leave the row without making any changes by clicking the "Esc" key. Then the editor highlights the row as "changed".
- Swap windows: Click this button to exchange windows. This is necessary if you compare two resources, as their order results from the order displayed in the table and the system does not know which resource is to be changed. If you only select one resource, this button is not available as in this case you can only change the optimized program version.
- Save: Save the changes made to the file on the left-hand side.

Processing notes for workplaces and machines

Configuration changes

Restart the terminal which the workplace/ machine is assigned to in order for the terminal program to interpret the configurations or modifications made to this workplace/machine.

Deleting a machine/ workplace

In a first step, the system shows a confirmation prompt asking if you really want to delete the machine. If you confirm this prompt, the system makes an attempt to delete the workplace. You can only delete a workplace successfully, if:

- you have not collected data for the workplace;
- you have not assigned the workplace to a terminal or a line;
- you have not logged on operations to the workplace;
- you have not planned operations for the workplace.

If you delete the workplace successfully, the system also deletes all configuration data, e.g. status assignments, for this workplace.

→←

Checking Business Parameter Containers (BSCs)

See [here](#) for further information on checking the system with respect to business parameters.

7 Status Texts

Summary

Menu	Master data → Workplaces/ machines → Status texts
Transaction code	Mstt
Function authorization	Mdmstt

In the status text dialog, all possible states (statuses) are designated with a descriptive text. These descriptions are then used in the [status assignment](#) performed later. The goal is to use standard status texts for all workplaces and machines.

Usage

In the status text dialog, all possible states (statuses) are designated with a descriptive text. These descriptions are then used in the [status assignment](#) performed later. The goal is to use standard status texts for all workplaces and machines.

Integration

During status assignment, a status text is assigned to a status.



The colors configured in the status text can be displayed in different evaluations/reports.

Selection criteria

The application provides the following selection criteria:

Status text

Unique status text number

Designation

Status text name

Field descriptions

Status text

Unique status text number

Designation

Informative description of the status. This text is displayed at the terminal, among other places.

Color

You also have the option to assign a color to a status text. This color is taken into account at various places in MOC.

Recommendation:



When setting up a new status text, the status number should preferably match the number of the status that will later be created during status assignment.

Example: The status text 2 "Set up" should be assigned to the status 2 during status assignment.

8 Status Assignment

Menu	Master data → Workplaces/ machines → Status assignment
Transaction code	mst
Function authorization	mdmst

Statuses can be created for all workplaces/ machines. The status illustrates the current state at the workplace/ machine.

Usage

All possible states (= statuses) or disturbances at a machine/ workplace are configured in status assignment and are assigned to status texts using unique status numbers. When malfunctions are detected at the terminal, the system falls back on the status recorded here.

Example of how a status table is set up

Status	1	2	3	4	5	6	12
Status text number	1	3	4	5	6	2	12
RPA	MUT	LCI	SCI	SET	BKS	DCI	DCI
Control for machine monitoring	Prod.	Disturbance	Disturbance	Disturbance	Disturbance	Disturbance	Gen. disturbance
Manual assignment at the terminal	☐	☐	☐	☒	☒	☐	☒
Automatic assignment via operating signal	☒	☒	☒	☒	☐	☒	☐
Input	1	2	3	4	0	6	0
Assignment at the MSS	0	2	1	3	-	7	-

The entries are based on the status texts listed below:

- 1 Production
- 2 Tool breakage
- 3 No raw material
- 4 Staff shortage
- 5 Setup
- 6 Break
- 12 Gen. malfunction

This is only an example to illustrate status assignment.

The selection dialog allows you to select and display statuses that have already been assigned to a machine or a workplace.

Integration

The workplace/ machine status can be found in several different evaluations.

Requirement

The following objects must be created before workplace/ machine statuses can be assigned.

- Workplace/ machine
- Status text
- Status class (optional)
- Resource performance account

Toolbar



Status list

Clicking on the "Status list" icon opens the report showing the defined and highlighted statuses and displays the statuses in clear text and as a barcode in the print-ready report.

Detailed help can be found under the following link: [machine status list report](#).

Selection criteria

The application provides the following selection criteria:

Workplace

The option to select the workplace/ machine the statuses of which you would like to display.

Status

The option to select the defined workplace/ machine statuses. The statuses are displayed for all workplaces/ machines with the status number entered.

Status text

Selection based on defined status text. You can also run a search using wildcards.

RPA abbreviation

Abbreviation for the assigned resource performance account

Status class

Abbreviation for the assigned status class

Field descriptions



Please note with regard to status 30000

Only selected options can be modified for status 30000:

- Warning in the machinery
- Activate production lock
- Activate machine lock
- Recording scrap reason (scrap reasons)

Certain fields such as status class, RPA and production identifier are not filled in for this status and therefore the fields are empty in the list.

General index tab

Machine/ short name

By providing the machine number, the state of the malfunction is assigned to a machine. In order to create a complete status table for a machine, each of the states of this machine must show the same number as what is input in this field.

The short name contains the designation of the machine entered or selected.

Status

A unique number is issued in this field for the status in the status table. This is the same number by which the state can be assigned or modified at the terminal.



Only one status may be defined at workplaces of the type "group workplace". This workplace has to be assigned the control indicator "production" (see "control" tab).

Please note: The status 30000 "Not assigned" cannot be deleted; furthermore, only selected fields can be configured for this status.

Superior status

If no status is defined in the superior status field, then this relates to a status of the uppermost level.

Otherwise, the number entered here is the one for the status directly above it. This status must already exist and be provided with a "hierarchy level" control indicator.

Please note: This function is available only on Windows terminals

Status text

The number assigned here refers to the clear text status message from the status text table.

Status class

By assigning a status to a status class, cumulative evaluations/reports can be performed on status classes. The abbreviation assigned here refers to the clear text status class message from the status class table.

Resource performance account

The status is assigned to a resource performance account (RPA) based on the entry in this field. You select one of the 12 RPAs for this.

The 12 RPAs are already predefined in HYDRA standard. The meanings for each of the RPAs are found in the glossary.

Control index tab

Production identifier

The following identifiers are available to control machine monitoring. Except for the "production" control indicator, all following control indicators are only allowed for machines/workplaces of the type "individual workplace".

Production

The production state/ status of a machine/ workplace is identified with Production. For each machine/ workplace, there must be **exactly one** status marked with this identifier. The HYDRA machine data collection function uses this identifier to detect the status that is automatically activated during production phases for machines/workplaces of the type "individual workplace". Only one status is allowed for "group workplaces". This status is to be assigned this control indicator.

Other status

All statuses that are not assigned one of the control indicators described here are to be provided with the control indicator "other status". As many statuses as required can be identified by this indicator per individual workplace.

General disturbance

One status per machine/ workplace **must** be created as General disturbance and be given this identifier. If the HYDRA machine data collection function detects a production phase that has not yet been assigned a disturbance or status, then this duration is posted to the general disturbance.

Material change

This option is only available with the MPL module.

Materials that are not planned can also be logged on for configurable statuses. They are logged on as alternative material, which means that the operator posts some material instead of the planned material. The determination whether or not alternative materials may be logged on for components is made in status assignment. As a result, the configuration must contain at least the status "Setup" in order to ensure the desired posting behavior. The machine status "Material change" option makes this posting possible as early as during operation logon without having to first perform a status change.

No order

When an operation is manually logged off or interrupted, a check is run at the Windows terminal to verify whether this is the last operation at the workplace. If no more operations are logged on at the terminal for this workplace, the terminal sets the status of this workplace to "No order".



The "No order" option is only available at Windows terminals (CTWIN/AIP). The option may only be set at individual workplaces, not at group workplaces, because only the "Production" status is supported at group workplaces. A maximum of one status may exist with this production identifier.

Short-term status (as of MDE 7.2)

For an optimized overview, for example in the status log or machine history, one status per machine can be configured as short-term disturbance. The purpose of this status is to provide a "container" for unconfirmed statuses that were active for only a specific (short) time during machine monitoring.

If a downtime is automatically detected at the terminal and the machine goes back into production automatically, the system verifies if the duration of this disturbance is shorter than the duration configured for this machine as short-term disturbances.

If it is, then the disturbance showing no reasons up until now is set with a reason using the status configured at the machine as the status for "short-term disturbances".

Such automatic status postings (reasons) for short-term disturbances are displayed within HYDRA in exactly the same manner as if they had been given a reason by an operator. During an automatic shift change, the duration of short-time disturbances is ignored.

Hierarchy level

Statuses that cannot be recorded at the terminal, and are therefore only to be used to display the hierarchy, are identified as "hierarchy level".



This function is only available at the Windows-based terminals; in this case, nothing more can be entered in certain configurations and index tabs.

Estimated downtime (as of MDE 7.2)

Once a status is set, an estimated downtime can be entered during manual status assignment; in the process, the downtime defined in the master data is shown as a suggestion.

During automatic status change, the downtime defined in the master data is automatically set.

Activate production lock (only for terminal types CT 56x, CT 7xx and CT 8xx)

If this option is set, production lock (P lock) is automatically activated at the terminal during status assignment.

During machine monitoring, this production lock prevents the status "Production" from being automatically set because of an incoming production signal. Consequently, the status set here overrides the production signal until the production lock is manually canceled at the terminal. Furthermore, the production lock can also affect whether and how the quantities accumulated during this time (counter readings) are posted.

Set machine lock output

This option must be set if machine lock should be activated during status assignment.

If it is, you must also ensure that machine lock output is configured accordingly in the machine configuration.

Status change index tab

Manually at the terminal

If this option is activated, the operator can enter this status manually at the terminal (by barcode or keyboard). If this option is not activated, the status will also no longer be displayed in the status selection list at the terminal.

Authorization

Access authorization for entering a status at the terminal (a value from 0 to 9). An authorization level for machine status modification is defined for every person in the HR master data. If the authorization level in the master data is lower than the authorization defined here, the operator cannot assign this status at the terminal.

Automatically via digital input

This option must be activated if a status assignment is made automatically via the machine interface (CT-MSS, CT-UMPS, PCC). The number of the digital input through which a status occurrence is detected must also be entered in the "input" field (0 = no input).

When a machine is monitored via an operating signal the operating signal is also acquired via a digital input. A status must be defined as an operating signal in the following form:

- Control for machine monitoring: Production
- Automatically assigned via operating signal: activated
- Input > 0 is entered. This input is used as operating signal.

As far as disturbance reasons are concerned, we generally distinguish between automatically and manually entered malfunction reasons. The disturbance reason entered manually at the terminal overrides automatically entered malfunction reasons.



If no production lock is set, the status with the control indicator "Production" overrides every malfunction reason. Therefore, keep in mind that the status "Production" must by all means be deactivated if an active malfunction is to be processed via an input.

If multiple automatic statuses (disturbance reasons) are active via digital inputs, the status with the lowest HYDRA channel number (not the lowest status number!) is set.



Please keep in mind that how the number is assigned to a physical connection to the MSS depends on the settings in the local terminal configuration file (Windows terminal: CTWIN.INI/CTAIP.INI, DOS terminal: AIOP.CFG).

Processing index tab

Log staff off

When this option is set, all persons currently logged on are logged off when a status is assigned (useful, for example, during maintenance phases). Otherwise, the people stay logged on.

Operation posting

An automatic overhead cost order logon can be activated during a status change using the option **Operation posting**. The following options are supported:

None

No processing.

Interrupt OP

This setting causes all active operations to be automatically interrupted and all people to be logged off at the workplace when switching to this machine status.

Interrupt active OP and log the following OP on

When status is changed, all active operations are interrupted and all people are logged off. The "subsequent operation" defined in the field *Operation* is automatically logged on.

Please note: The subsequent operation may not be subject to batch management.

Transfer registered person to OP

Depending on this setting, people are transferred to the "subsequent operation" defined in the field *Operation*.

If the operation is an overhead cost operation or if the posting workplace is a group workplace (GAP), then at least one person is logged on to the subsequent OP: either the person posting the operation, or if there is no field available to enter a staff badge number in the change status dialog, this will be the person logged on the longest. If no one is logged on at this time, the subsequent operation cannot be logged on.

Scrap reason

Automatically recorded scrap can be posted to a defined reason, depending on the relevant active status.

A distinction is made in the process, whether the production lock is set or not while the status is active. This reflects the configuration, which provides for either a scrap reason while the status is active and also while the status with a production lock is active.

A counter input explicitly defined as scrap reason counter generally has a higher priority than a reason defined here.

Plausibilities index tab

Check for running operation

Statuses can be configured so that

- | | |
|---|--|
| - they are only allowed if an OP is logged on to a workplace | Option <i>An operation must be logged on</i> |
| - they are only allowed if no OP is logged on at a workplace | Option <i>No operation may be logged on</i> |
| - they are allowed irrespective of the presence of an OP at a workplace | Option <i>No check</i> |

HYDRA checks this dependency during status change as well as during logon/ logoff or interruption of operations. If the condition is violated by the desired posting, the terminal displays an error message and rejects the posting.



If, when an operation is terminated or interrupted, a new status is set, this status may not be an order-related status!

Use of unplanned material allowed

This option becomes relevant when using HYDRA material and production logistics (MPL). If the option is set, the use of unplanned material is allowed, i.e. batches that are not specified in the input material list at the operation can be logged on as input material. This can be useful, for example, during setup.

User fields index tab**User field key**

This field is preset with the value DEFAULT and as a rule it cannot be modified. MPDV defines the user fields for this user field key during the customizing process to meet specific customer requirements.

9 Setting Outputs Depending on Status and Posting Scenario

Usage

Under advanced configurations, it is possible to set the machine's digital outputs to account for when a certain machine status and a defined posting scenario coincide.

The following combination of machine status and posting scenario can be configured:

- A status is active without a production lock
- A status is active with a production lock
- A status is active, but no operation is logged on
- A status is active and at least one operation is logged on
- A status is active, but no person is logged on
- A status is active and at least one person is logged on

If the situation changes (status modification, activate or deactivate P lock, OP posting, personnel posting), the terminal checks the logical configured outputs and sets them as necessary.

In doing so, there is the option to set exactly the same logical output for several statuses within one machine as well as several statuses of different machines.

If a logical output is set for different conditions at the same status, then it will be set if one of the conditions is met (OR link).

Requirements

- This function is only available at the Windows CTWIN or AIP terminals.
- For this purpose, the terminal must be configured in operation mode "MDE".
- The machine must be permanently assigned to the "MDE terminal".
- Output signals that are used for this function may not be configured anywhere else at the same time (e.g. machine lock). This must be assured during configuration. This is not checked automatically.
- The terminal must be restarted any time a change is made to the configuration of the logical outputs.
- The outputs configured here may not be used simultaneously with the "inputs/ outputs" configuration that exists in the [resource configurations](#).

Procedure

The configurations required for this are defined in the so-called "[object-related configuration](#)".

Object type	fix "MSTAT" (Machine STATus)
ID1	Machine number for a numeric machine number, always add leading zeros to make it eight digits
ID2	Machine status number
ID3	(leave empty)
ID4	(leave empty)
Parameters	The following parameters are supported: MST_NO_PLOCK Output/ signal for "Status active without production lock" MST_PLOCK Output/ signal for "Status active with production lock" MST_NO_ANR Output/ signal for "A status is active, but no operation is logged on" MST_WITH_ANR Output/ signal for "A status is active and (at least) one operation is logged on" MST_NO_PNR Output/ signal for "A status is active, but no person is logged on" MST_WITH_PNR Output/ signal for "A status is active and (at least) one person is logged on"
Parameter value	The parameter value is always the number of the logical output that is to be set.
Active	Fix "J"

Example

Output 1 is to be set at machine 100 if either setup (status 2) or dismantling (status 9) is active with a production lock.

Output 2 is to be set at machine 100 if production (status 1) is active although no operation is logged on.

Object type	ID 1	ID 2	ID 3	ID 4	Parameters	Parameter value	Active
MSTAT	100	2			MST_PLOCK	1	Y
MSTAT	100	9			MST_PLOCK	1	Y
MSTAT	100	1			MST_NO_ANR	2	Y

10 Print Status List

Summary

Menu	Master data → Workplaces/ machine status assignment → Status list
Transaction code	- (mst for machines/ workplace status assignment)
Function authorization	mdmst.print

Usage

The purpose of this function is to generate a status list with clear text and barcodes. A barcode provides the ability to enter the status at a terminal using a barcode reader (e.g. a scanner gun).

Integration

The function is called up from the status assignment of machines/ workplaces.

A printed barcode provides the ability to enter the status at a terminal using a barcode reader (e.g. a scanner gun).

To print a status list, follow the steps listed below:

1. In the list, highlight the status you would like to print out on the status list. Click on the top left-side corner of the table if you want to highlight all statuses.
2. Click on the "status list" icon to call up the print preview screen. In it, there will be one new page for each workplace/ machine.
3. Click on the icon "Print report"  in the print preview to print out the status lists on the default printer set in MOC.

Structure of the workplace/ machine status barcode

NNNNN0

Place	Designation	Length
*	Asterisk	1
N	Machine status, with preceding zeros	5
0	fixed: 0	1
*	Asterisk	1
Length of status barcode without asterisks		6

Please note:

By default, HYDRA supports the barcodes "39", "128" and "Interleaved 2 of 5"

The system only supports barcode detection and automatic assignment to the corresponding input fields at the terminal for barcode readers that are connected at the serial port (COM port). This is not possible for barcode readers that are "looped in" through the keyboard.

11 Cycle Parameters

Summary

Menu	Master data → Workplaces/ machines → Cycle parameters
Transaction code	cycpa
Function authorization	mdcycl

Usage

HYDRA provides the ability to monitor cycle times within the machine data recording function without requiring HYDRA process data processing to be used.

The purpose of this function is to configure the action and tolerance limits.

Integration

Values are marked in different colors in the *Workplace overview* depending on whether the action or tolerance limit was exceeded:

Standard	Black
If the value drops below or exceeds the action limit	Blue
If the value drops below or exceeds the tolerance limit	Red



If a limit is exceeded, no further processing steps are taken in HYDRA.

Requirement

Before defining any configurations, you must first set up the [machine](#).

Selection criteria

The application provides the following selection criteria:

Machine

Selection by machine/ workplace

Field descriptions

Machine

Machine for which the configuration applies.

Tolerance limit positive, negative

Values may not drop below or exceed the percentage values defined here. The cycle time of the logged on operation is always used as the target value for cycle time monitoring. This can be corrected at the terminal. The limit value is entered as a percentage of the target value.

Example: Target value: 20 sec/ cycle
tolerance positive: 10 %
tolerance negative: 5 %

Thus, this results in the following limit values:

Upper limit value: 22 sec/ cycle
lower limit value: 19 sec/ cycle

Action limit positive, negative

Percentage values can be entered here, triggering a warning once they have been reached. This is why the action limits should be defined more narrowly than the tolerance limits. The limit value is entered as a percentage of the target value.

12 Counter Configuration

Summary

Menu	Master data → Workplaces/ machines → Counter configuration
Transaction code	ctr
Function authorization	mdctr

Usage

Counter configuration defines the behavior of how the pulses transmitted by MSS (machine interface) should be interpreted in terms of monitoring and quantity.

Integration

In addition to defining counter configuration, each time a machine is connected to the terminal several different settings must be configured.

Requirement

Before defining any configurations, you must first set up the [machine](#).



Configuration changes

In order for the shop floor terminal program to be able to interpret the settings or changes that were made, you must first restart the terminal that is assigned to the workplace/ machine.

Selection criteria

The application provides the following selection criteria:

Machine

The ability to select the defined counter for the chosen resource/ machine.

Field descriptions

Machine

By entering the machine number, a logical counter channel is assigned to a machine.

Counter input

Counter number used to uniquely identify the counter channel for the machine.

Please keep in mind that the counter number assignment to the physical connection at the CT-MSS, CT-UMPS or OPC variable depends on other settings as well (e.g. local terminal configuration file CTWIN.INI/CTAIP.INI (Windows terminal) or AIOP.CFG (DOS terminal)).

Designation

Explanation of the counter, e.g. "Yield at closure counter"

Unit

Quantity unit that is recorded by the counter. Is currently only used for documentation purposes.

Cycles monitoring

For channels where this identifier is set, incoming counter pulses are considered during cycle monitoring.

When calculating the actual cycle, only the first counter is referenced that was configured using the identifier "For monitoring". This counter must also be set for monitoring via operating signal if an actual cycle is to be calculated at the same time.

Posting as

Defines which quality of the counter quantity is recorded. Possible values:

- Yield
- Scrap
- Rework
- Open quantity

As a rule, quantity postings are booked to the primary quantities account.

Counters with the option "No posting" do not post to a quantity account. Only the counter reading is recorded. Furthermore, a posting can be made as cycles (strokes) and/or be controlled using cycle monitoring.

Reason

When posting quantities for scrap, rework or open quantities, a reason can be stored at the counter. To do so, you must first create the reason in the configuration accordingly (*Reasons*).

Thus, for example, you can define two separate counters with different scrap reasons.



A scrap reason defined within counter configuration has a higher priority than a scrap reason defined at the status.

Posting as cycles

If this option has been set, the pulses coming in via this counter are posted as cycles.

Allocation with partitioning

If this option is set, the incoming pulses are compensated using the respective active partitioning/pulse factor.



For counters with the option "No posting", the setting "Posting with partitioning/ pulse factor" is irrelevant, because no cycles are partitioned into quantities for these counters.

Allocation with

The pulses recorded using this counter and the resulting quantities can be offset against a different quantity account. In this case, the recorded quantity is deducted from the account against which it should be offset.

Keep in mind that the result of relevant quantity compensation may also be a negative value.



What you must be aware of for DOS based terminals is that you must make sure you do NOT set the option "Offset against" in the machine/ workplace configuration, index tab "Quantities", if a compensation is set here.



Counters with the option "No posting" cannot be offset against quantity accounts.

Last editing

Editor, date and time of last editing

Notes for DOS based terminals

At the DOS based terminals (CT-56x, CT-541), the counter configuration is only supported to a limited extent:



Only a maximum of one counter can be configured for each quantity account "Yield" or for each quantity account "Scrap" (no reworking, no open quantity, no counter with the option "No posting").



An "Offset against" is only possible at the second counter that must be configured as "Scrap". In this case, the compensation is only possible with "Yield".

What you must be aware of is that you must make sure you do NOT set the option "Offset against" in the machine/ workplace configuration, index tab "Quantities", if a compensation is set here.



No reason may be defined for the scrap counter.

Toolbar



Workplace configuration

Starts the [configuration of workplaces/resources](#).

13 Terminal Assignment

Summary

Menu	Master data → Work places → Terminal assignment
Transaction code	mta
Function authorization	mdmta

Usage

This function is used to configure the machine assignment to the terminal.

Usage

A machine must be assigned to a terminal in order to be able to use the MDE specific functions such as automatic shift change, cyclical status update or data entry via MSS (machine interface). However, these functions are only available if the terminal is configured as the so-called "MDE terminal" (see terminal configuration). These functions are not active for terminals that are configured as ADE terminals, not even if the machines/ workplaces are assigned to the terminal.

An assignment of this kind makes it so that the machine/workplace is displayed by default at the terminal.

The number of assignable machines depends on the terminal type that is specified.

Terminal type CT-541:	Only one assignment possible
Terminal type CT-76x, CT 83x, CT84x:	Up to 16 assignments possible (even if the terminal is configured as the master terminal) Up to ten machines can be used for process data processing (PDV).
Terminal type CT-56x:	Up to 8 assignments possible
Terminal type A-SUB	Up to 20 assignments possible.

All of the machines that have been assigned to terminals up until now appear based on the selected terminals. The order displayed on the terminal is the result of the position specified here.

If a line (only available if an MDE-LIN license was issued) is assigned to a terminal, all aggregates of the line are automatically assigned to the terminal and displayed in gray under the position "99". Aggregates may not be removed from the assignment. If one line is removed from the assignment, the aggregates assigned to the line are automatically removed as well. Lines may only be assigned to terminal types CT76x, CT83x (two lines max) and CT84x (three lines max.).

Using the option button "Processing", the machine type assignment to the terminal can be differentiated. The following options are available:

A - ADE processing

M - MDE processing

Processing according to operation mode of terminal

Thus, both workplaces/ machines with MDE processing as well as workplaces with ADE processing only can be assigned to an MDE terminal.

For a group workplace, processing must be set to "ADE processing" if this is to be assigned to an MDE terminal.



Configuration settings

In order for the shop floor terminal program to be able to interpret the settings or changes that were made, you must first restart the terminal.

Selection criteria

The application provides the following selection criteria:

From to

Used to select the terminal number

Field descriptions

Terminal

Unique terminal number that a machine should be assigned to.

Position

The machine's display position at the terminal and in the terminal assignment at the client

Machine

Machine that the terminal was assigned to/ should be assigned to

Processing

Here, the following two options are available to choose from:

A ADE processing

M MDE processing

Processing as per operation mode of terminal

14 Postings Relating to Workplaces/Machines

Summary

Menu	Data collection --> Corrections --> Postings relating to machines
Transaction code	mboo
Function authorization	mboo

Utilization

This application enables the user to obtain an overview of the postings relating to workplaces/machines generated in HYDRA and to edit, record and correct relevant data quickly and easily.

The below-mentioned documents provide further information on postings relating to machines:

- [Posting of quantities](#)
- [Posting of Times](#)

Please note: The terms "posting" and "log record" are used synonymously in this document.

Integration

A log record always refers to a period of time in which a workplace/machine status is available. Each status change results in a new log record.



If the shift automatic function is enabled, the shift change also leads to a new log record, even if the status does not change in this case. If the status 30000 "not assigned" is available when shifts change, this period is posted onto the status with the control indicator "general disturbance".

Prerequisite

Machine status changes need to be recorded for the workplaces/machines.

Selection criteria

The application provides the following selection criteria:

Workplace ... to ...

Determines the machine-related postings that were posted onto the specified workplace/machine.

Group ... to ...

Determines the machine-related postings that were booked onto workplaces/machines of the entered group (according to the resource configuration).

Cost center

Determines the machine-related postings that were booked onto workplaces/machines of the entered cost center (according to the resource configuration).

Company

Determines the machine-related postings that were posted onto workplaces/machines of the entered company (according to the resource configuration).

Responsibility area

Determines the machine-related postings that were booked onto workplaces/machines of the entered responsibility area (according to the resource configuration).

Date from ... to ...

Determines machine-related postings the start time of which is within the specified period of time. In this context, the period is either defined in connection with specifying the shift or the time.



Please note that this application only allows for data of the online data area to be selected and edited.

Shift

Restricts the date range defined above by selecting by the shift. Example: if the date range is selected from 23 January 2009 until 25 January 2009 and shift 1 is chosen, all machine-related postings the start time of which is in shift 1 of 23 January 2009, shift 1 of 24 January 2009 or shift 1 of 25 January 2009 are selected.

Time

Restricts the date range defined above by selecting by the time. Example: if the date range is selected from 23 January 2009 until 25 January 2009 and the time from 3 pm until 11 pm is chosen, all machine-related postings the start time of which is between 3 pm on 23 January and 11 pm of 25 January 2009 are selected..

If several selection criteria are used, the results matching these criteria will be displayed.

Field Descriptions

“General” tab

Record type

P: This log record was generated by a status change

N: This log record was generated at the end of a shift

Confirmed/uploaded

This flag cannot be changed. It is set within the scope of customer-specific uploads.

Machine

Machine/workplace for which the log record was generated.

Cost center

Cost center of the machine/workplace according to the workplace configuration for which the log record was generated.

Posting start, posting end

Period of time (beginning, end) for which the log record applies.

Duration

Duration since the last status change

Shift

Shift (number) for which the log record applies.

Beginning of shift

Beginning of the shift for which the log record applies.

End of shift

End time of the shift for which the log record applies.

Status

Status which coincided with the period of time of the log record.

The resource performance account is determined automatically from status assignment.

Comment

Comment (max. 60 characters) for the machine status change, if entered while assigning statuses (dialog M_MST) on the terminal.

Target cycle

Target cycle that was set when the status was changed (end of log record).

Partitioning

Partitioning that was set when the status was changed (end of log record).

Editor

Editor who modified the log record manually the last time.

Last modification

Point in time when the log record was edited manually the last time.

“Quantities“ tab**Yield (P)**

Computed yield in primary quantity unit for counter values that were recorded during the period of the log record or yield in primary quantity unit that was manually recorded during the period of the log record.

Scrap (P)

Computed scrap in primary quantity unit for counter values that were recorded during the period of the log record or scrap in primary quantity unit that was manually recorded during the period of the log record.

Rework (P)

Computed rework quantity in primary quantity unit for counter values that were recorded during the period of the log record or rework quantity in primary quantity unit that was manually recorded during the period of the log record.

Open quantity (P)

Computed open quantity in primary quantity unit for counter values that were recorded during the period of the log record or open quantity in primary quantity unit that was manually recorded during the period of the log record.

Unit (P)

Primary quantity unit according to a configuration relating to workplaces/machines (resource configuration).

Yield (S), Scrap (S), Rework (S), Open quantity (S), Unit (S)

The corresponding quantities in secondary quantity unit are converted from the primary quantity according to the definitions made for conversions in the workplace configuration and provided that corresponding conversion factors are available.



Please consider the "convert quantities" option for manual quantity changes.

Yield (T), Scrap (T), Rework (T), Open quantity (T), Unit (T)

The corresponding quantities in tertiary quantity unit are converted from the primary quantity according to the definitions made for conversions in the workplace configuration and provided that corresponding conversion factors are available.

Please consider the "convert quantities" option for manual quantity changes.

Yield (B), Scrap (B), Rework (B), Open quantity (B), Unit (B)

The corresponding quantities in base quantity unit are converted from the primary quantity according to the definitions made for conversions in the workplace configuration and provided that corresponding conversion factors are available.

Please consider the "convert quantities" option for manual quantity changes.

Convert quantities

When quantities are changed, the "convert quantities" option allows for quantities to be converted according to the conversion factors defined at the machine or the configured formulas for quantity conversion.

If the conversion factors of active operations are to be used for the machine (compare the option "basis for HYDRA-MDE quantity conversion" in the workplace configuration), it is urgently recommended not to activate the "quantity conversion" option, when it comes to subsequent changes in the maintenance of postings function of HYDRA-MDE, as only conversion factors that are specific to machines are taken into account by the maintenance of postings function.

Cycles

The number of cycles that were recorded within the period of the log record.

Toolbar



Postings relating to orders

Function authorization: oboo.*

Starts the function "[Order-related postings](#)".

Transfers the workplace, cost center, date/time and shift from the selection area.